

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM

101/2

MATHEMATICS

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Instructions to candidates

- Write your **name**, **admission** number and **school** in the spaces provided above.
- Sign and write the date of the examination in the spaces provided above.
- The paper contains **TWO** Sections: **Section I** and **Section II**.
- Answer **ALL** the questions in Section I and **only five** questions from **Section II**.
- All answers and working must be written on the question paper in the spaces provided below each question.
- Show all the steps in your calculations, giving your answers at each stage in the spaces below each question.
- Marks may be given for correct working even if the answer is wrong.
- Non-programmable** silent electronic calculators and KNEC Mathematical tables may be used except where stated otherwise.

For Examiner's Use Only

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL
17	18	19	20	21	22	23	24	TOTAL								

SECTION II

SECTION I (50 marks)

Answer **all** questions in this section in the spaces provided

1. Simplify without using mathematical tables or calculator.

(4mks)

$$2(\log_{10} 2.5 + \log_{10} 40)$$

$$3 \log_{10} 0.05 + 2 \log_{10} 2 - \log_{10} 0.5$$

2. Simplify $\frac{2}{2+\sqrt{5}} - \frac{2\sqrt{5}}{2-\sqrt{5}}$ and express your answer in the form $a+b\sqrt{c}$ where a, b and c are constants.

(3mks)

3. A wedding committee did a budget for a wedding ceremony as follows:

Food: Ksh. 58,205

Chairs: Ksh. 11,950

Entertainment: 8,453

The sum of the budget was done by first rounding each figure to 3 significant figures.

a) Determine the sum of the **budget**

(2mks)

b) Determine the percentage error in this sum of the budget

(2mks)

4. Solve the equation $4\sin^2x + 4\cos x = 5$ for $0^\circ \leq x \leq 360^\circ$

(3mks)

5. a) Expand $(1-x)^4$ using the binomial expansion

(1mk)

b). Use the first three terms of the expansion in (a) above to find the value of $(0.998)^4$

Correct to the nearest hundredth

(3mks)

6. Make w the subject of the formula

(3mks)

$$P = \sqrt{\frac{vw^2}{V^2 - W^2}}$$

7. Given that $y = 3 \sin \left(\frac{1}{2}x + 60 \right)^0$ find, amplitude, period and the phase angle of the function.

(3mks)

8. A ship sails due North from latitude 20^0 S for a distance of 1440nm. Find the latitude of the point it reaches

(2mks)

9. The equation of a circle is given by $3x^2+3y^2+3x+42y+30=0$. Determine the radius and the coordinates of the centre circle.

(3mks)

10. a) i) Draw a straight line MN such that $MN=7\text{cm}$

(1mk)

- ii) Construct the locus P such that $\angle MPN=90^\circ$

(1mk)

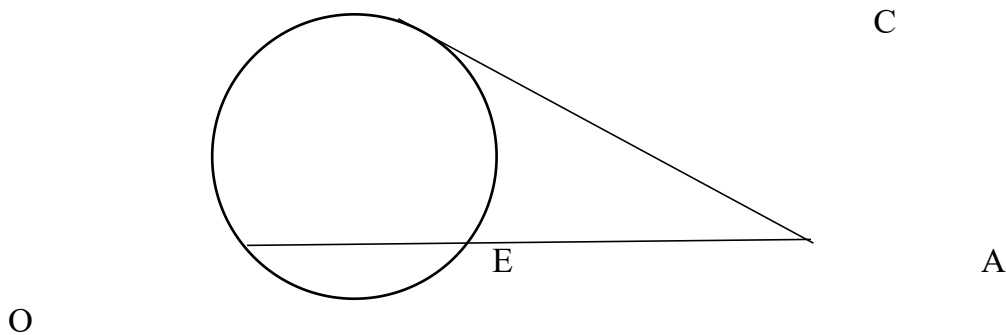
- b) On the locus of P in (a) above, mark point T such that T is equidistant from M and N. **(2mks)**

11. The table below shows tax table for monthly income

Monthly taxable income in Ksh.	Tax rate % in each shilling
0- 9680	10
9681- 18800	15
18801-27920	20

In a certain month, Kamau's tax was sh. 3336. Determine his income during that month. **(3mks)**

12. In the figure below OC is the tangent to the circle. If OE=8cm and OC=6cm, find EA. **(2mks)**



13. Evaluate $\int_1^4 (3x^2 + 1)dx$
(3mks)

14. Liquid P contains 30% of water while liquid Q contains 48% of water. In what ratio should P and Q be mixed so that the mixture contains 42% of water? (3mks)

15. The probability that it is rainy in the morning is 0.6. The probability that John carries an umbrella while going to work is 0.4. Find the probability that

i) It is not rainy and John does not carry an umbrella. (2mks)

ii) It is rainy and John carries an umbrella
(1mk)

16. Solve the simultaneous equations

$$x-2y=1, \quad x^2+y^2=29$$

(3mks)

SECTION II (50 MARKS)

(Answer five questions in this section)

17. The first three consecutive terms of a geometric progression are 3^{2x+1} , 9^x and 81 respectively

a) Calculate the value of x

(3mks)

b) Find the common ratio of the series

(2mks)

c) Calculate the sum of the first ten terms of this series

(2mks)

d) Given that the fifth and the seventh terms of this Geometrical Progression form the first two consecutive terms of an arithmetic sequence calculate the sum of the first 20 terms of the arithmetic sequence.

(3mks)

18. In an experiment, the length of 100 rats were measured to the nearest 0.1cm and the frequency tabulated as follows:

Length in (cm)	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59	60-64
Frequency	1	4	x	20	25	26	7	1	1

a) Find the value of x (2mks)

b) Calculate the mean length using assumed mean of 42
(4mks)

c) Calculate the standard deviation
(4mks)

19. Complete the table below for the function

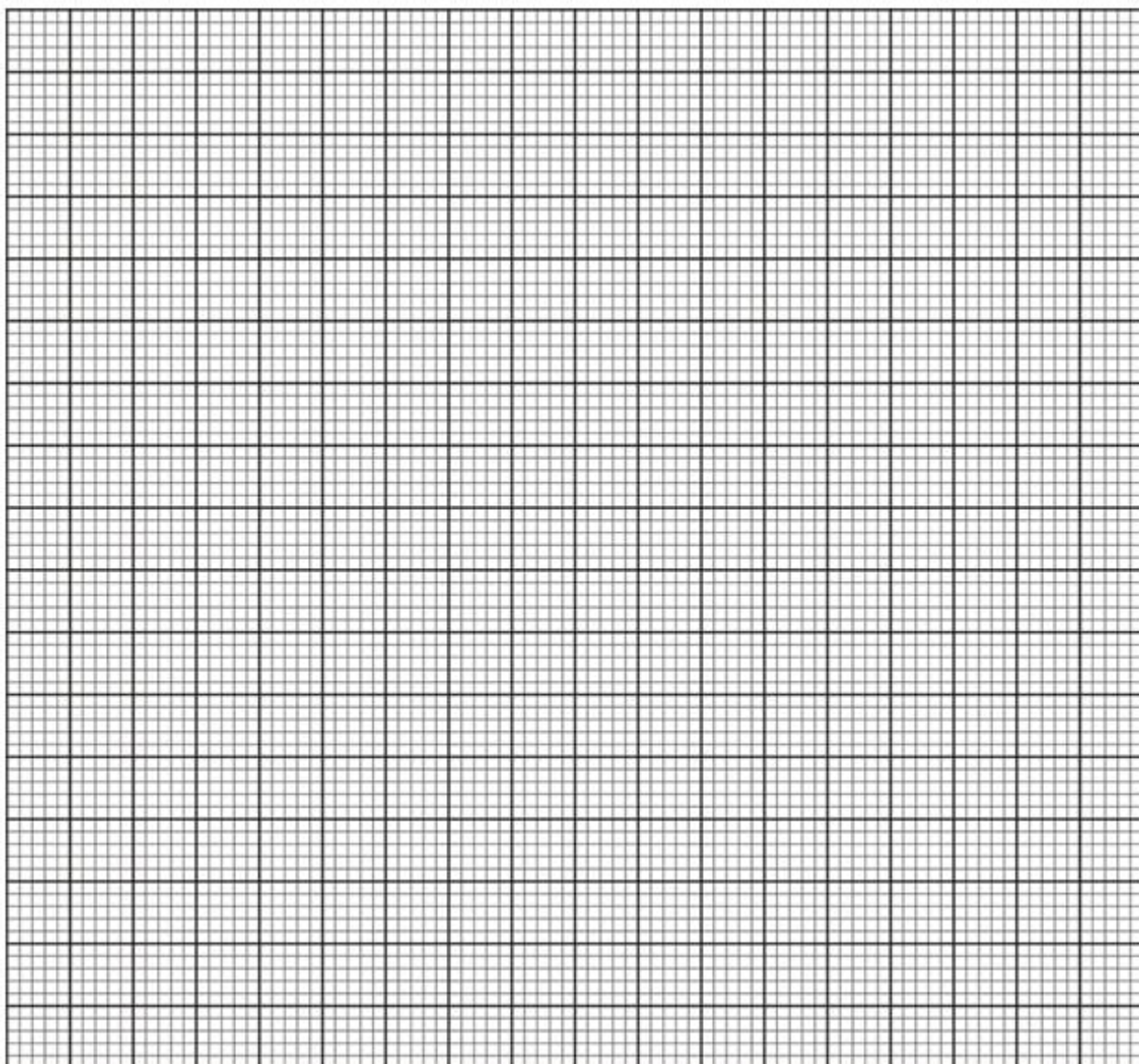
$$y=x^3+6x^2+8x$$

(2mks)

X	-5	-4	-3	-2	-1	0	1
x^3	-125	-64		-8	-1	0	1
$6x^2$		96	54		6	0	6
$8x$	-40		-24	-16		0	8
Y			3	0	-3	0	15

a) Draw the graph of the function $y=x^3+6x^2+8x$ for $-5 \leq x \leq 1$ (3mks)

(Use a scale of 2 big squares to represent 1 unit on the x -axis and 1 big square to represent 2 units on the y -axis)



b) Use your graph to estimate the roots of the equations

i. $x^3 + 6x^2 + 8x = 0$

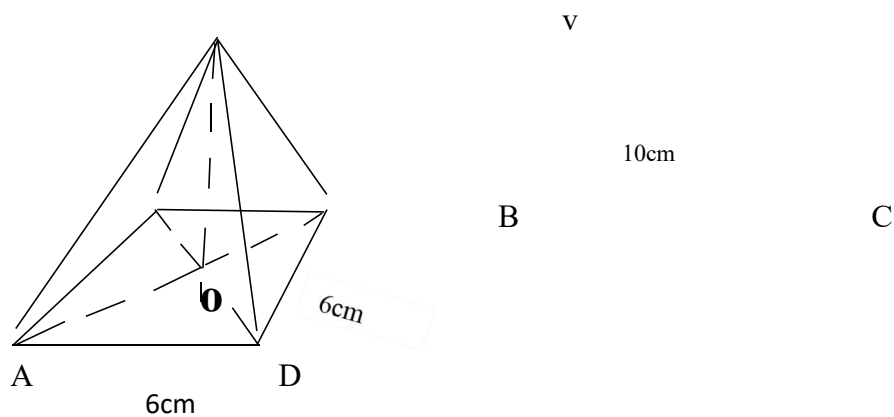
(1mk)

ii. $x^3 + 5x^2 + 4x = -x^2 - 3x - 1$

(2mks)

c) Find the values of x which will satisfy the inequality $x^3 + 6x^2 + 8x > 1$ **(2mks)**

20. The figure below is a square based pyramid ABCDV with $AD=DC=6\text{cm}$. $VO=10\text{cm}$



a) State the projection of VA on the base ABCD

(1mk)

b) Find:

i. The length of VA

(3mks)

ii. The angle between the planes VA and ABCD

(2mks)

iii. The angle between the planes VDC and ABCD

(2mks)

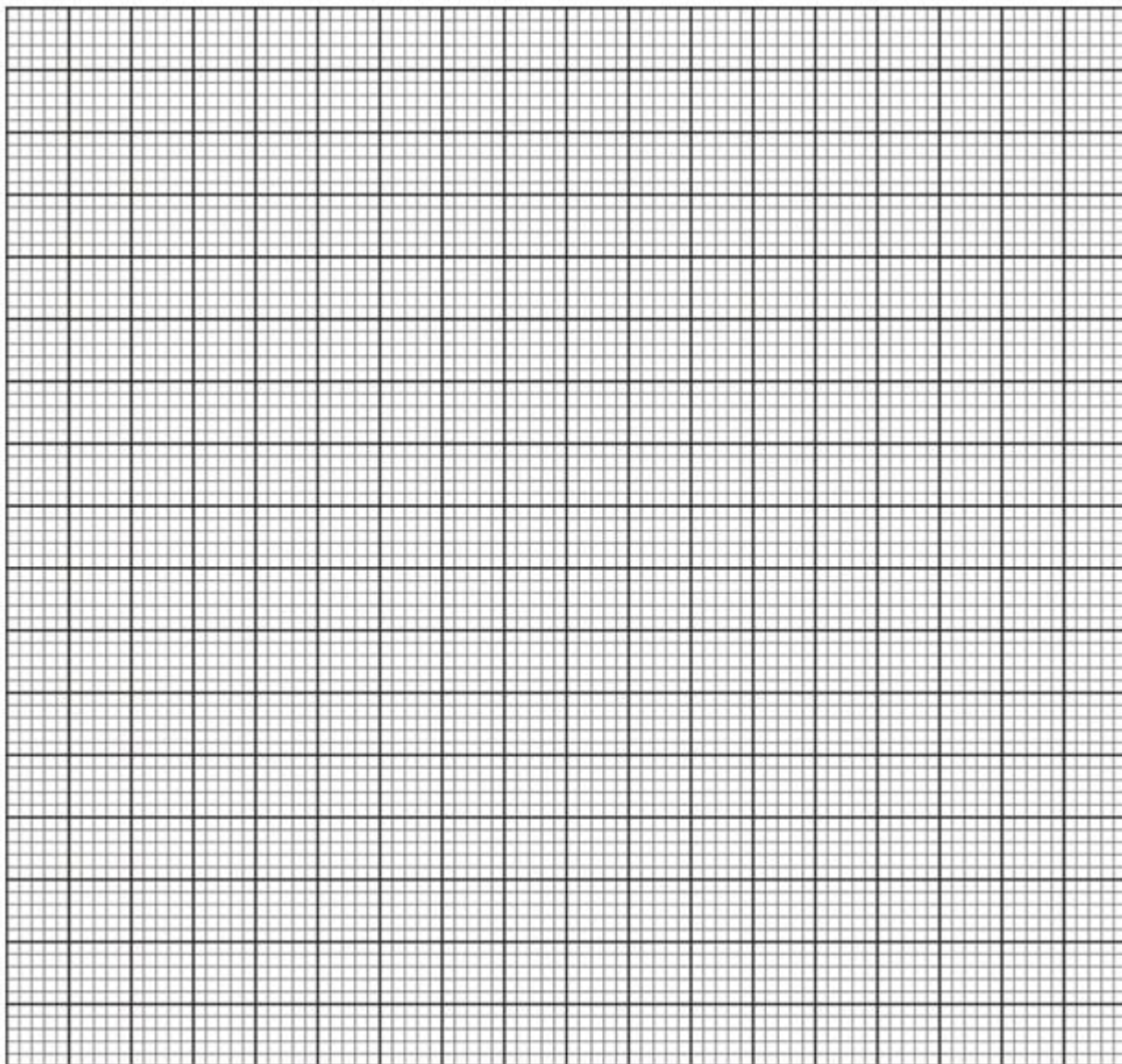
iv. The volume of the pyramid

(2mks)

21. The points A (0,0), B (-3,1), C (1,3) and (4,2) are the vertices of a parallelogram ABCD.

a) A'B'C'D' is the image of ABCD under the matrix of transformation $\begin{bmatrix} & \\ & \end{bmatrix} \begin{matrix} -2 & 0 \\ 0 & -2 \end{matrix}$

Draw ABCD and A'B'C'D' on the grid. Write down the coordinates of A'B'C'D' **(3mks)**



- b)** The points $A''(0,0)$ $B''(-6,2)$ $C''(2,6)$ and $D''(8,4)$ are the vertices of $A''B''C''D''$ the image of ABCD under a certain transformation. Draw $A''B''C''D''$ on the same grid as ABCD. Describe this transformation fully. **(3mks)**

c) A single transformation T maps $A'B'C'D'$ on to $A''B''C''D''$. Determine the matrix of T **(4mks)**

22. A sum of money is deposited in a bank that pays simple interest at a rate r . After 3 years the total amount of money in the account is Ksh. 358,400. The interest earned each year is Ksh. 12,800.

a) Calculate: i) The amount of money which was deposited.

(2mks)

ii) The rate of interest r .

(2mks)

b) A computer whose marked price is Ksh. 40,000 is sold at ksh. 56,000 on hire purchase terms.

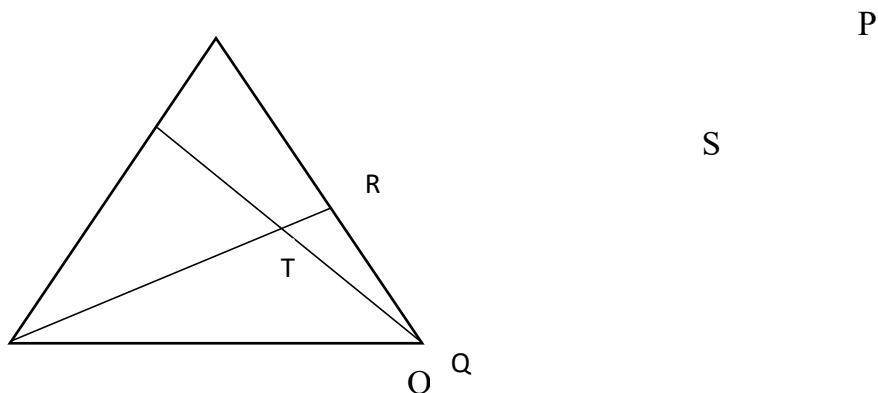
i) James bought the computer on hire purchase terms. He paid a deposit of 25% of the hire purchase price and cleared the balance by equal monthly instalments of ksh. 2625. Calculate the number of instalments.

(3mks)

ii) Had James bought the computer on cash price he would have been allowed a discount of 12.5% on the marked price. Calculate the difference between the cash price and the hire purchase price and express it as a percentage of the cash price.

(3mks)

23. In the figure below OPQ is a triangle in which $\vec{OS} = \frac{3}{4} \vec{OP}$ and $\vec{PR} : \vec{RQ} = 2:1$. Lines OR and SQ meet at T.



- a) Given that $\vec{OP} = \mathbf{p}$ and $\vec{OQ} = \mathbf{q}$. Express the following vectors in terms of $\mathbf{p}$ and $\mathbf{q}$:

i. $\vec{PQ}$

(1mk)

ii. $\vec{OR}$

(2mks)

iii. $\vec{SQ}$

(2mks)

- b) Given that $\vec{ST} = m\vec{SQ}$ and $\vec{OT} = n\vec{OR}$ where m and n are consonants, determine the values of m and n

(4mks)

c) Find the ratio of ST:TQ

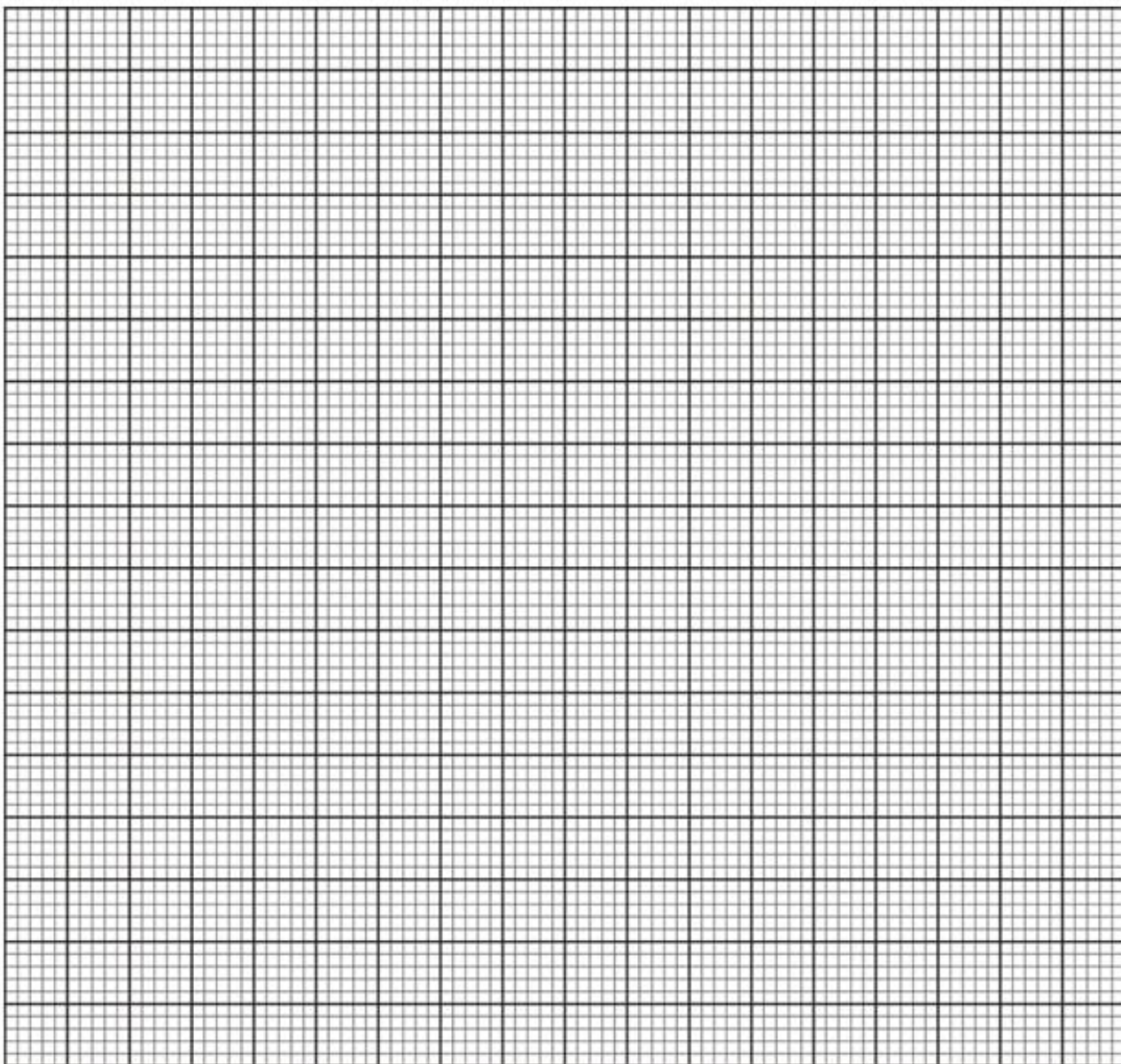
(1mk)

24. During installation of electricity bulbs in street lighting a dealer is required to supply two types of bulbs A and B. The total number of bulbs should not be more than 400. He must supply more of type A than of type B and type A should not be more than 300 and type B should not be less than 80.

a) By letting the number of type A bulbs to be x and the number of type B bulbs to be y , write all the inequalities representing the above information. **(3mks)**

b) On the grid provided draw all the **inequalities**

(4mks)



- c) If type A bulbs cost sh.450 per piece and type B cost sh.350 per piece and that the higher the cost the higher the profit:
- i. Use your graph to determine the number in each type of bulb that he should supply to maximize the profit.

(1mk)

- ii.** Calculate the maximum cost of lighting the street
(2mks)